

RAVI KUMAR D

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GitHub | LinkedIn

PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of experience designing, developing, training, and deploying machine learning and deep learning models in production. Expertise in computer vision, generative AI, foundation models, LLM integration, and end-to-end ML pipelines. Proficient in model optimization, inference deployment, and cloud-native AI architecture using AWS and Terraform.

PROFESSIONAL EXPERIENCE

Ava Software Private Limited | AI/ML Engineer | August 2025 – Present

Poubelle.ai (PROD) – Intelligent Waste Classification System

- Developed end-to-end ML pipeline for waste classification using YOLO achieving 88% accuracy with Amazon Bedrock fallback for continuous learning.
- Engineered automated monthly retraining pipeline with data augmentation improving accuracy by 6%, implementing model versioning and evaluation metrics.
- Built intelligent recommendation engine providing bin classification (recycle, compost, donate, landfill) with 91% user acceptance rate.

Thryve Labs – AI Virtual Assistant (PROD)

- Developed intelligent conversational AI system using LLMs for natural language understanding and multi-intent classification.
- Architected Strands agent-based API orchestration system managing 40+ client-provided endpoints, intelligently routing requests based on user intent with dynamic parallel or sequential invocation strategy.
- Implemented Redis-based response caching with 1-hour TTL for identical intents, reducing API latency by 70% and backend load by 45% through intelligent response reuse.
- Designed cloud-native infrastructure on AWS leveraging Amazon Bedrock for foundation model serving with auto-scaling and fault tolerance.

Shoob Photography (PROD) – Intelligent Background Removal & Image Enhancement

- Trained MODNet semantic segmentation model using transfer learning achieving 96% background removal accuracy.
- Engineered computer vision pipeline with PIL-based enhancement and edge artifact refinement for pixel-perfect results.
- Built scalable batch processing system supporting 100+ concurrent requests with AWS Lambda and S3 integration.
- Optimized inference through quantization reducing latency by 40% and costs by 35%.

Video Proctoring Engine (POC) – AI-Based Exam Monitoring System

- Engineered exam surveillance system using computer vision and audio classification detecting cheating patterns and suspicious activities.
- Developed facial analysis module with MediaPipe detecting eye movements and face orientation; audio classifier categorizing voice patterns with 94% accuracy.

- Integrated YOLO v5 object detection for unauthorized material identification during exam sessions.

Chaotic – AI-Powered Image Enhancement System (POC)

- Built generative AI system using fine-tuned Qwen (20B) foundation model for high-resolution image synthesis.
- Implemented prompt engineering and data augmentation pipeline for dynamic character modification with visual consistency.
- Engineered AWS S3 storage managing multi-TB datasets with optimized access patterns and sub-second inference latency.

Hema's Enterprise Private Limited | Backend Engineer | December 2024 – July 2025

CK Ticketing Tool – Workflow Management System

- Architected workflow system with dynamic forms, multi-level approvals, and automated email notifications.
- Designed optimized MySQL schema for analytics and reporting.

Luminar – Role-Based Project Management Platform

- Developed platform with role-based access control, task assignment, and real-time tracking.
- Integrated AWS S3 and OAuth 2.0; automated synchronization using CRON jobs.

TECHNICAL SKILLS

Machine Learning & Deep Learning: YOLO, MODNet, Semantic Segmentation, Computer Vision, Object Detection, Image Classification, Audio Classification, Convolutional Neural Networks (CNN), Transfer Learning, Model Fine-tuning, Hyperparameter Tuning, Model Evaluation, Inference Optimization, Quantization, Batch Processing, Model Serving

Generative AI & LLMs: Foundation Models, Large Language Models, Prompt Engineering, Intent Classification, Natural Language Understanding, Semantic Analysis, Amazon Bedrock, Qwen

Frameworks & Libraries: TensorFlow, scikit-learn, OpenCV, PIL, MediaPipe, NumPy, Pandas, PyTorch

Programming Languages: Python, PHP

Web Frameworks & APIs: FastAPI, Laravel, RESTful APIs

Cloud & MLOps: AWS (EC2, S3, Lambda, Bedrock), Terraform, Infrastructure Automation, Model Deployment, Auto-scaling

Databases & Data Processing: MySQL, PostgreSQL, SQL

Tools & Platforms: GitHub, Git, Visual Studio Code, Postman, Jira, Linux, Ubuntu, Windows

EDUCATION

Master of Science in Information Technology (M.Sc. IT)

St. Joseph's College of Arts & Science (Autonomous), Cuddalore | 2023 – 2025 | GPA: 80%

Bachelor of Computer Applications (B.C.A)

St. Joseph's College of Arts & Science (Autonomous), Cuddalore | 2020 – 2023 | GPA: 79%